

Autodesk® Scaleform®

Scaleform 最佳優化指南

Scaleform 4.0

Matthew Doyle
3.0
2011 4 25

Copyright Notice

Autodesk® Scaleform® 4.4

© 2014 Autodesk, Inc. All rights reserved. Except as otherwise permitted by Autodesk, Inc., this publication, or parts thereof, may not be reproduced in any form, by any method, for any purpose.

Certain materials included in this publication are reprinted with the permission of the copyright holder.

The following are registered trademarks or trademarks of Autodesk, Inc., and/or its subsidiaries and/or affiliates in the USA and other countries: 123D, 3ds Max, Algor, Alias, AliasStudio, ATC, AutoCAD LT, AutoCAD, Autodesk, the Autodesk logo, Autodesk 123D, Autodesk CAM 360, Autodesk Homestyler, Autodesk Inventor, Autodesk MapGuide, Autodesk Streamline, AutoLISP, AutoSketch, AutoSnap, AutoTrack, Backburner, Backdraft, Beast, BIM 360, Burn, Buzzsaw, CADmep, CAiCE, CAMduct, CFdesign, Civil 3D, Cleaner, Combustion, Communication Specification, Configurator 360™, Constructware, Content Explorer, Creative Bridge, Dancing Baby (image), DesignCenter, DesignKids, DesignStudio, Discreet, DWF, DWG, DWG (design/logo), DWG Extreme, DWG TrueConvert, DWG TrueView, DWGX, DXF, Ecotect, ESTmep, Evolver, FABmep, Face Robot, FBX, Fempro, Fire, Flame, Flare, Flint, FMDesktop, ForceEffect, FormIt, Freewheel, Fusion 360, Glue, Green Building Studio, Heidi, Homestyler, HumanIK, i-drop, ImageModeler, Incinerator, Inferno, InfraWorks, InfraWorks 360, Instructables, Instructables (stylized robot design/logo), Inventor, Inventor HSM, Inventor LT, Kynapse, Kynogon, LandXplorer, Lustre, MatchMover, Maya, Maya LT, Mechanical Desktop, MIMI, Mockup 360, Moldflow Plastics Advisers, Moldflow Plastics Insight, Moldflow, Moondust, MotionBuilder, Movimento, MPA (design/logo), MPA, MPI (design/logo), MPX (design/logo), MPX, Mudbox, Navisworks, ObjectARX, ObjectDBX, Opticore, Pipeplus, Pixlr, Pixlr-o-matic, Productstream, Publisher 360, RasterDWG, RealDWG, ReCap, ReCap 360, Remote, Revit LT, Revit, RiverCAD, Robot, Scaleform, Showcase, Showcase 360 ShowMotion, Sim 360, SketchBook, Smoke, Socialcam, Softimage, Sparks, SteeringWheels, Stitcher, Stone, StormNET, TinkerBox, ToolClip, Topobase, Toxik, TrustedDWG, T-Splines, ViewCube, Visual LISP, Visual, VRED, Wire, Wiretap, WiretapCentral, XSI.

All other brand names, product names or trademarks belong to their respective holders.

Disclaimer

THIS PUBLICATION AND THE INFORMATION CONTAINED HEREIN IS MADE AVAILABLE BY AUTODESK, INC. "AS IS." AUTODESK, INC. DISCLAIMS ALL WARRANTIES, EITHER EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE REGARDING THESE MATERIALS.

Autodesk Scaleform

Scaleform

Autodesk Scaleform Corporation
6305 Ivy Lane, Suite 310
Greenbelt, MD 20770, USA

www.scaleform.com

info@scaleform.com

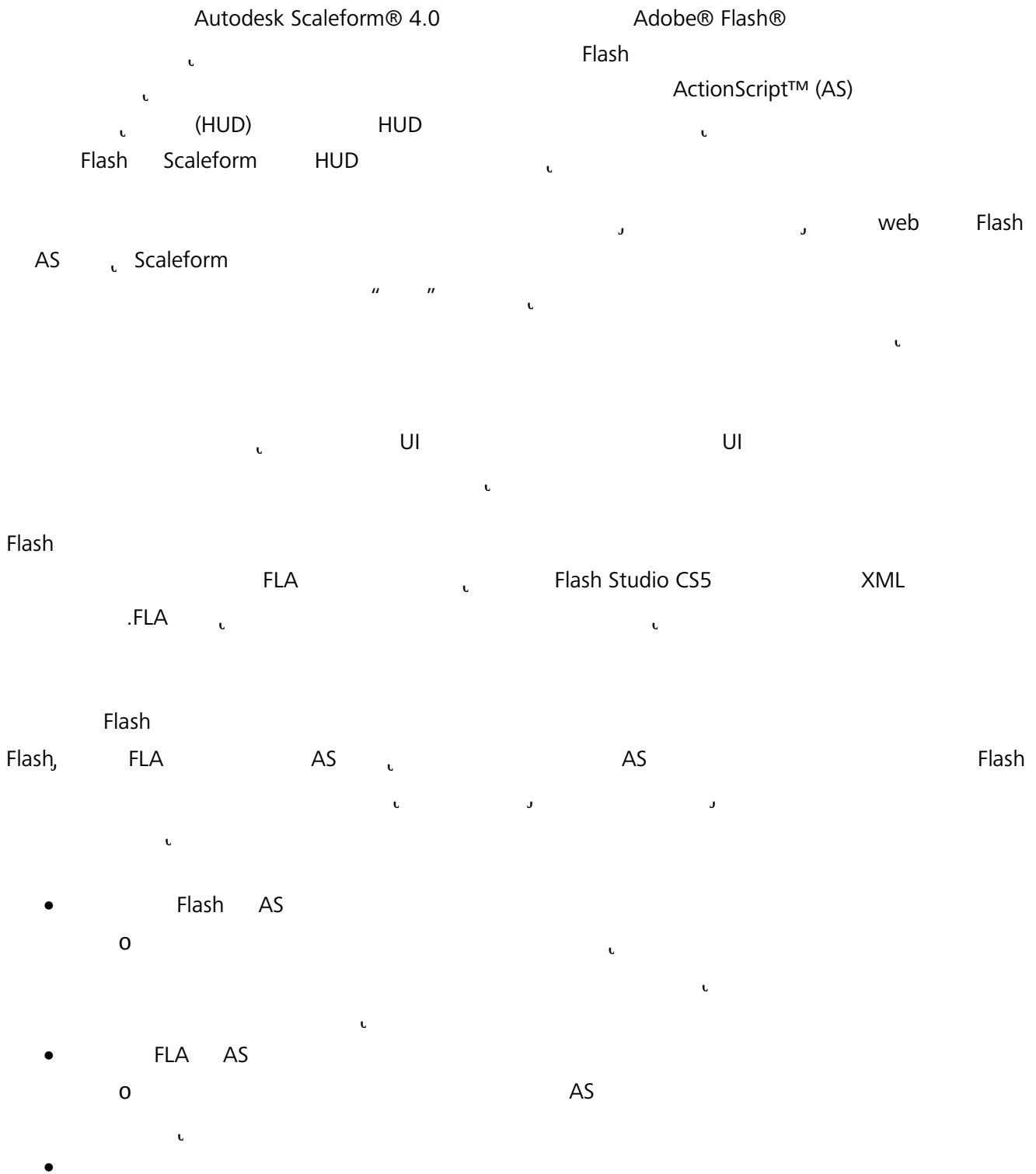
(301) 446-3200

(301) 446-3199

1	概要	1
1.1		2
1.2		2
2		4
2.1		4
2.2		4
2.3		4
2.3.1		5
2.3.2		5
2.3.3		6
2.3.4		8
2.3.5		9
3	ActionScript	11
3.1	ActionsScript2	11
3.1.1		13
3.1.2		13
3.1.3	/	14
3.2	一般 ActionScript 3 指导原则	14
3.2.1	严格数据类型确定	14
3.3		15
3.4	onEnterFrame and Event.ENTER_FRAME	15
3.4.1	清理 onEnterFrame	15
3.5	ActionScript 2 Optimizations	16
3.5.1	onClipEvent 和 on Events	16
3.5.2	Var 關鍵字	16
3.5.3	預存	17
3.5.4	預存長路徑	18
3.5.5	複雜運算式	18
4	HUD	20
4.1	SWF	20

4.2		SWF	21
4.3			22
4.4		(Advanced)	22
4.5	Flash	HUD	23
5			25
5.1	Flash		25
5.2			25
6			26

1 概要



0

-

Flash AS

0

Flash

Flash

Flash AS

Flash

-

-

ActionScript

Flash

Flash

1.1

-

-

UI

Flash

Scaleform 3.3

Scaleform

(CLIK™)

1.2

-

-

Flash

HUD

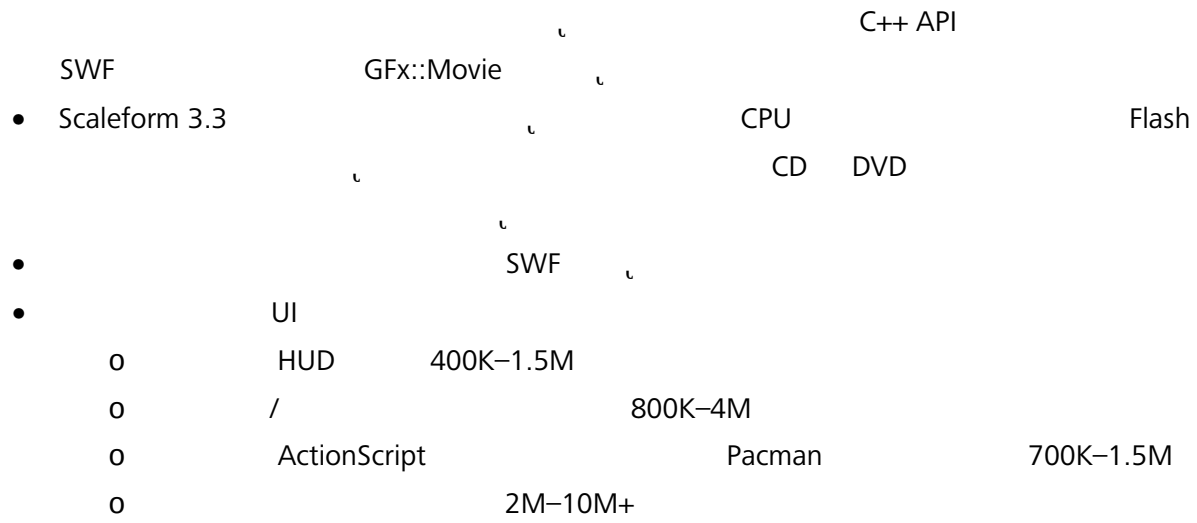
Flash

MMORPG

Flash

在 AS 端，不同文件可以通过 AS2 和

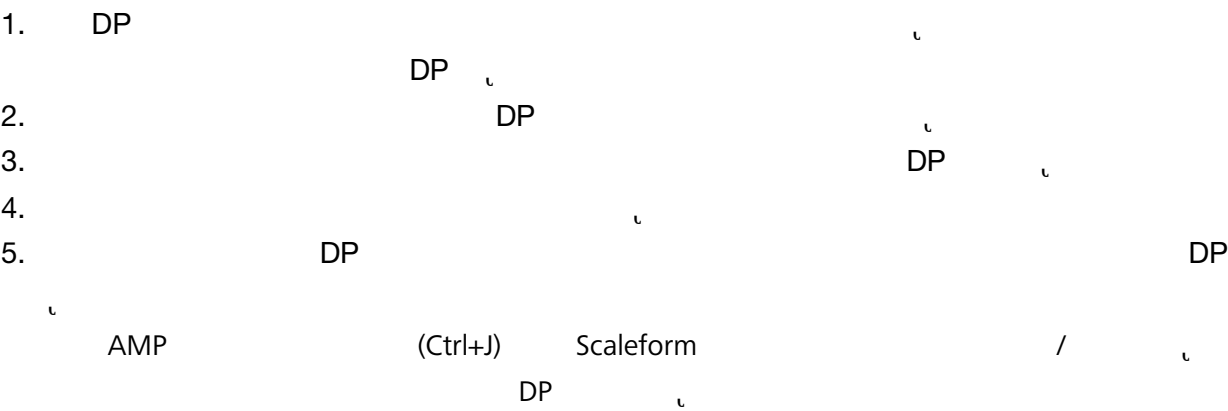
AS3 中的各种 SWF 加载 API 进行加载。



2

Flash

2.1



2.2

- 1.
- 2.
3. `_visible=false` and `visible=false` rather than `alpha=0` (in AS3). `_alpha=0_alpha=0` (in AS2) "stop() "

2.3

2.3.1

Flash

Scaleform

2.3.2

-
-

0
0

alpha Flash

`_visible`(AS2) or `visible` (AS3) `false`
`_alpha_alpha` (AS2) or `alpha` (AS3) 0, `_alpha alpha` property
100

Flash

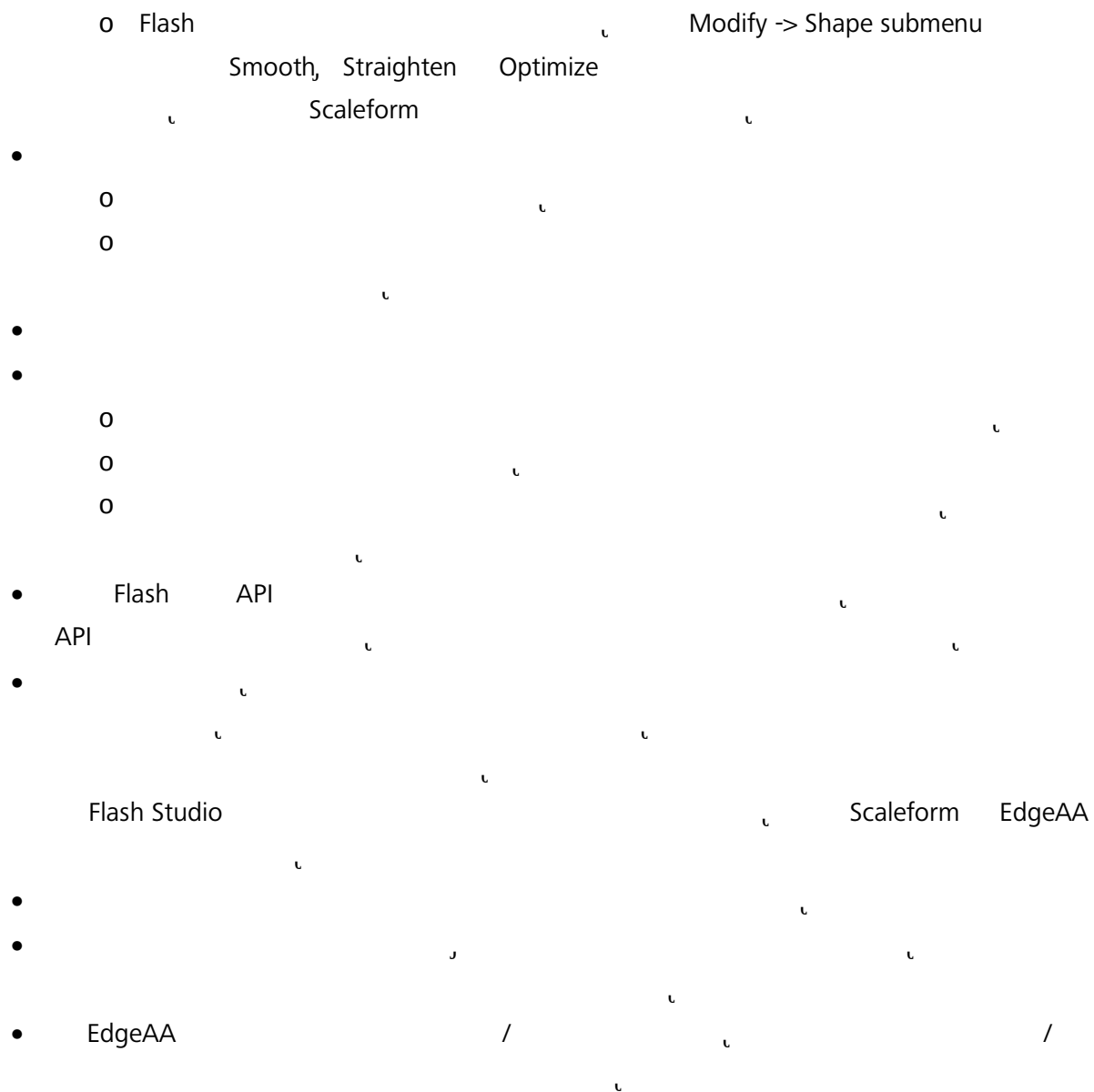
`_x` `_y`(in AS2)

or x and y properties (in AS3)

`_visible /visible` `false` Flash

-

0
0



2.3.3

Flash

SWF _visible false _alpha 0

Scaleform _alpha 0 false CPU ActionScript CPU

SWF false _visible

- / 2 16x32, 256x128,
- 1024x1024 512x32,
- JPEG JPEG
- DXT gfxexport SWF GFX
- DXT DDS gfxexport
- -qp -qh
- /
- Edge AA
- ActionScript
- SWF/GFX SWF 1024 x 1024 JPEG
- 4 MB
- UI gfxexport -i DDS 4
- AMP
- CPU
- gradient.swf SWF gfxexport
- -d0 gradients.swf SWF
- Adobe Photoshop® Flash
- PNG

-
-

Flash

2.3.4

FLA

SWF

AS

12 FPS 24 FPS 12 FPS 24 FPS 12 FPS 24 FPS 5

[onEnterFrame](#) [onEnterFrame](#) (in AS2) or [Event.ENTER_FRAME](#) (in AS3)

[onEnterFrame/Event.ENTER_FRAME](#)

[setInterval](#)

[onEnterFrame/Event.ENTER_FRAME](#)

[setInterval](#)

30 40 FPS
Flash UI

CPU

-
-

on/off

[_visible/visible = true/false](#)

/

-

tween

0

tween

tween

/

0

tween

Flash Tween

0	Scaleform	CLIK Tween	gfx.motion.Tween in AS2 or scaleform.clik.motion.Tween in AS3	Flash Tween
---	-----------	------------	--	-------------

```

graph LR
    morphing --> CPU
    tween --> CPU
    tween --> Flash
    60 --> Flash
    CPU --> Flash
    Flash --> 60FPS[60 FPS]
    60FPS --> Flash
  
```

2.3.5

- SlotHeight gfxexport 48 DP
- DP
-
-
-
- ("TextField.variable")
- "htmlText" HTML
- gfxexport -fc, -fcl, -fcm
- " "

-
- The diagram illustrates the relationships between various UI components and APIs. It includes labels such as 'DP', 'TextField', 'DrawText API', 'Flash', 'Scaleform', 'Flash UI', 'C++', and 'C+++'. Arrows indicate dependencies or relationships between these elements.

3 ActionScript

ActionScript



3.1 *ActionScript2*

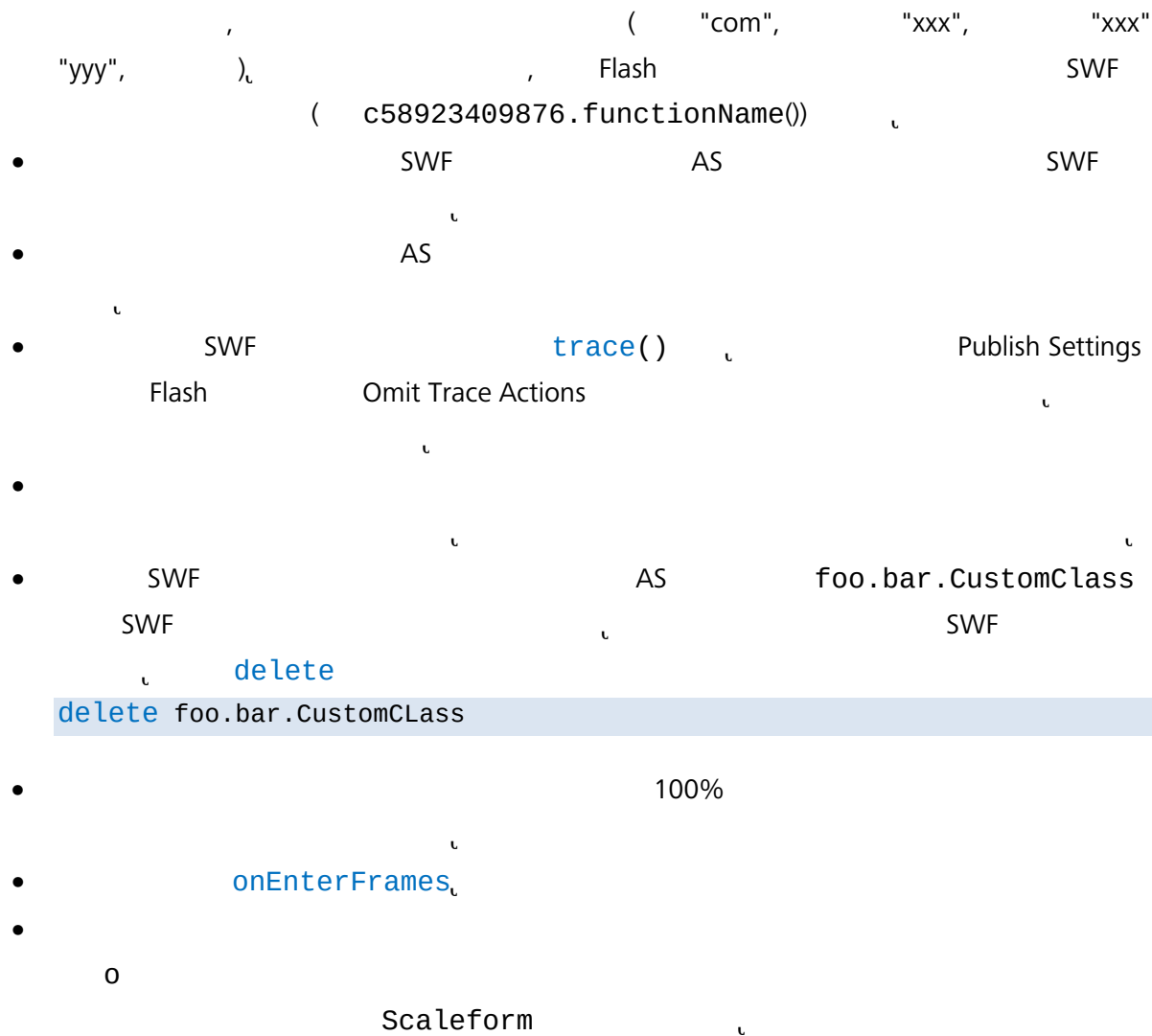
- AS
 - attachMovie FLA
- AS
 - on() onClipEvent() onEnterFrame, onPress
 - “ if ”
 - Flash AS
 - (onClipEvent, on)
 - (gotoAndPlay, play, stop)
- 避免在包含多個動畫的長時間軸電影剪接中使用“遠距”gotoAndPlay/gotoAndStop。
 - gotoAndPlay/gotoAndStop,
 - ,gotoAndPlay/gotoAndStop
 - gotoAndPlay/gotoAndStop

- 0 gotoAndPlay/gotoAndStop, ,
gotoAndPlay/gotoAndStop
- gotoAndPlay/gotoAndStop
gotoAndPlay/gotoAndStop
- gotoAndPlay/gotoAndStop
C++
- [Gfx::Movie::SetVariableArray](#) C++ AS
- Scaleform
Scaleform
- 0 bug
- eval()
- Array.length myArr.length

```
var fontArr:Array = TextField.getFontList();
var arrayLen:Number = fontArr.length;
for (var i:Number = 0; i < arrayLen; i++) {
    trace(fontArr[i]);
}
```

```
var fontArr:Array = TextField.getFontList();
for (var i:Number = 0; i < fontArr.length; i++) {
    trace(fontArr[i]);
}
```

- null
- removeListener()
- AS VM
- AS "if"
- "com.xxx.yyy.aaa.bbb", VM "com", "xxx", "yyy", "aaa", "bbb",
"if" / /



3.1.1



3.1.2

-
- `with`

3.1.3 /

-
- “ var ” “ var ” ActionScript
-
-
- `null`
- SWF `null`
-
- AS

3.2 一般 *ActionScript 3* 指导原则

AS3

3.2.1 严格数据类型确定

-
- (Object) AS3 “
-
- (Array) (Vector) 4294967295 /
- `Vector<*>`
- `Vector<*>` `Vector<int>`, `Vector<String>` `Vector<YourFavoriteClass>` `Vector<*>`
- `flash.utils.Dictionary`
- `GfX` Flash
- / AS2 AS3
- AS3

3.3

1. AS `onEnterFrame/ Event.ENTER_FRAME`
2. UI
3. `(_visible(AS2) or visible (AS3) true`
`stop()` Advance
4. `_global.noInvisibleAdvance(AS2) or`
`scaleform.gfx.Extensions.noInvisibleAdvance (AS3)`
Advance “true”
- Flash Flash
`_global.gfxExtensions true` Scaleform
- 5.
- 6.

3.4 *onEnterFrame and Event.ENTER_FRAME*

- `onEnterFrame(AS2) or Event.ENTER_FRAME (AS3)`
- `onEnterFrame/Event.ENTER_FRAME`
- `setInterval setTimeout setInterval`
- `clearInterval`
- `setInterval setTimeout onEnterFrame/Event.ENTER_FRAME`
`onEnterFrame/Event.ENTER_FRAME`

3.4.1 清理 `onEnterFrame`

`delete onEnterFrame`

```
delete this.onEnterFrame;
delete mc.onEnterFrame;
```

```
    null    undefined    onEnterFrame(    this.onEnterFrame = null;)
    onEnterFrame    Scaleform
onEnterFrame
```

3.5 ActionScript 2 Optimizations

3.5.1 onClipEvent 和 on Events

onClipEvent() on() onEnterFrame, onPress

-
- onClipEvent
this, _global, arguments, super 256

onLoad
onClipEvent(construct) onEnterFrame

```
onClipEvent(construct)
{
    this.onLoad = function()
    {
        //
    }
}
```

onClipEvent(load)

3.5.2 Var 關鍵字

var AS
var AS

```
var i = 1000;
countIt = function()
{
    num = 0;
```

```

    for(j=0; j<i; j++)
    {
        j++;
        num += Math.random();
    }
    displayNumber.text = num;
}

```

```

var i = 1000;
countIt = function()
{
    var num = 0;
    var ii = i;
    for(var j=0; j<ii; j++)
    {
        j++;
        num += Math.random();
    }
    displayNumber.text = num;
}

```

3.5.3 預存

var

```

function foo(var obj:Object)
{
    for (var i = 0; i < obj.size; i++)
    {
        obj.value[i] = obj.num1 * obj.num2;
    }
}

```

```

function foo(var obj:Object)
{
    var sz = obj.size;
    var n1 = obj.num1;
    var n2 = obj.num1;
}

```

```

    for (var i = 0; i < sz; i++)
    {
        obj.value[i] = n1*n2;
    }
}

```

```

var floor = Math.floor
var ceil = Math.ceil
num = floor(x) - ceil(y);

```

```

var keyDown = Key.isDown;
var keyLeft = Key.LEFT;
if (keyDown(keyLeft))
{
    //
}

```

3.5.4 預存長路徑

```

mc.ch1.hc3.djf3.jd9._x = 233;
mc.ch1.hc3.djf3._x = 455;

```

```

var djf3 = mc.ch1.hc3.djf3;
djf3._x = 455;

var jd9 = djf3.jd9;
jd9._x = 223;

```

3.5.5 複雜運算式

```

, C
this[_global.mynames[queue]][_global.slots[i]].gosplash.text =
_global.MyStrings[queue];

```

```

var _splqueue = this[_global.mynames[queue]];
var _splstring = _global.MyStrings[queue];
var slot_i = _global.slots[i];
_splqueue[slot_i].gosplash.text = _splstring;

```

```

for(i=0; i<3; i++)
{
    this[_global.mynames[queue]][_global.slots[i]].gosplash.text =
                                                _global.MyStrings[queue];
    this[_global.mynames[queue]][_global.slots[i]].gosplash2.text =
                                                _global.MyStrings[queue];
    this[_global.mynames[queue]][_global.slots[i]].gosplash2.textColor =
                                                0x000000;
}

```

```

var _splqueue = this[_global.mynames[queue]];
var _splstring = _global.MyStrings[queue];
for (var i=0; i<3; i++)
{
    var slot_i = _global.slots[i];
    _splqueue[slot_i].gosplash.text = _splstring;
    _splqueue[slot_i].gosplash2.text = splstring;
    _splqueue[slot_i].gosplash2.textColor = 0x000000;
}

```

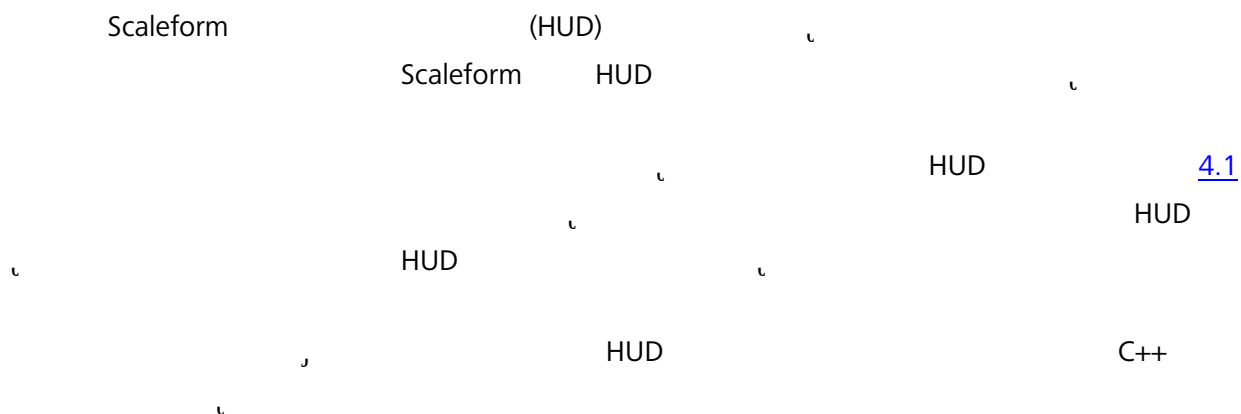
6

```

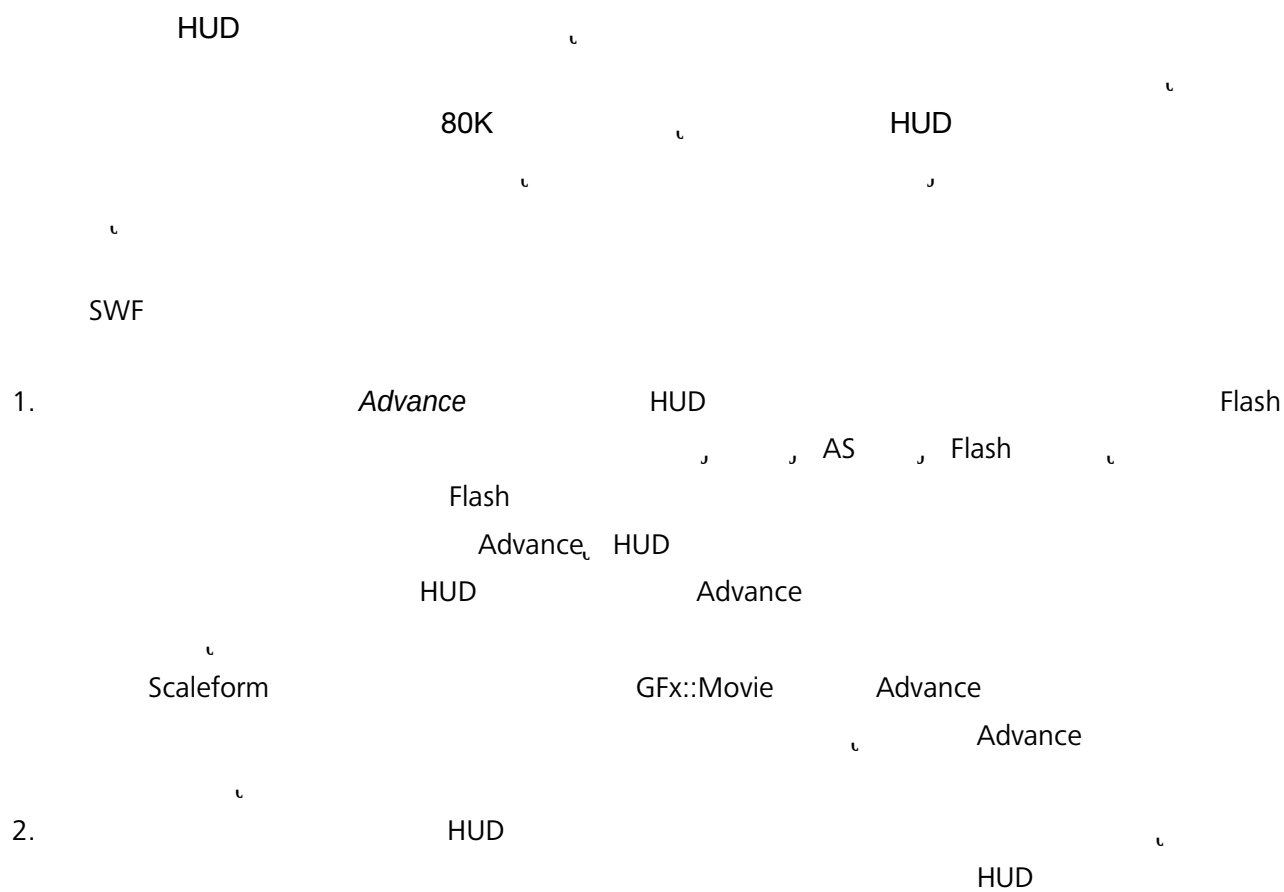
var _splqueue = this[_global.mynames[queue]];
var _splstring = _global.MyStrings[queue];
for (var i=0; i<3; i++)
{
    var slot_i = _global.slots[i];
    var elem = _splqueue[slot_i];
    elem.gosplash.text = _splstring;
    var gspl2 = elem.gosplash2;
    gspl2.text = splstring;
    gspl2.textColor = 0x000000;
}

```


4 HUD



4.1 SWF



4.2

SWF

Flash AS `loadMovie`

Flash

1. `_visible = false` (AS2) or `visible = false` (AS3)

`_global.noInvisibleAdvance` (AS2) or
`scaleform.gfx.Extensions.noInvisibleAdvance` (AS3) `true`

HUD

`_visible/visible = false`

Advance

Flash

HUD

HUD

Advance

`_global.gfxExtensions` (AS2) or `scaleform.gfx.Extensions.enabled` (AS3)

`true` Scaleform

2. `SetVariableArray`

`SetVariableArray`

3. `onEnterFrame` (AS2) or `Event.ENTER_FRAME` (AS3)

HUD `onEnterFrame//Event.ENTER_FRAME`

Advance

4. AS

Flash

30-40 FPS
HUD

AS
15-20 FPS

	Flash	Flash
Flash	4.2	



٢

L

1

1

•

“ _____ **”**

FAQ₆

0

23

•

u

u

•

u

2.1

" _____ "

u

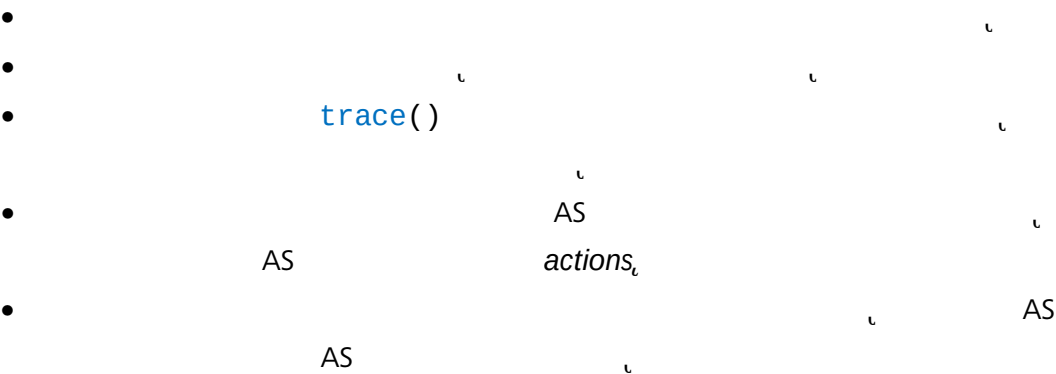
FAQ_u

5

5.1 Flash



5.2



6

ActionScript 2.0

http://www.adobe.com/devnet/flash/articles/as_bestpractices.html

Flash 8

http://www.adobe.com/devnet/flash/articles/flash8_bestpractices.html

Flash ActionScript 2.0

http://www.adobe.com/devnet/flash/articles/actionscript_guide.html